

“Learning on the Job”

How Summer Internships Complement School-Based Learning for Computer Science Students

Mora Labisi · Eleanor O’Rourke — Computer Science & Learning Sciences, Northwestern University

RESEARCH QUESTION

What do computing students learn during their summer internships, and how does it compare to their school-based learning?

MOTIVATION

Internships as an approach to bridging the academia-industry gap

STUDY

We conducted **semi-structured interviews** with six CS students, during and immediately after their internships*, and **inductively coded** their responses.

*Due to a scheduling conflict, Cleo was interviewed only once, after her internship concluded

6
interns

11
interviews

2
timepoints

FINDINGS

INTERNSHIPS WERE SITES OF AUTHENTIC CS LEARNING (Tran et al., 2025)

Professional Relevance

Many participants used terms such as “**real**” and “**actual**” to describe and recount the technical work they did during their internships.

Legitimate Mentorship

Participants spoke about being surrounded by **industry professionals** and the various benefits (and drawbacks) that came with it.

Personal Agency

Participant descriptions of their internship work, especially as opposed to their schoolwork, were **open-ended and self-directed**.

Broader Impacts

The internship experience helped participants gain discernment and acquire a (dis)taste for **future courses and career paths**.

PARTICIPANTS LEARNED...

• **TECHNICAL SKILLS** • **PROFESSIONAL PRACTICES** • **INTRICACIES OF THE WORKPLACE**

Agentic AI

Coding with AI

Debugging

Navigating an Existing Codebase

Version Control / GitHub

Adapting Explanations

Documentation Habits

Planning Before Writing Code

Self-Directed Work

Thorough Code Review

Working in Ambiguity

Navigating Mismatched Expectations

(Human) Networking

Time Management

Working in a Company Culture That's Not a Good Fit

ALL SIX PARTICIPANTS LEARNED SOMETHING SCHOOL DOES NOT TEACH

"I'm constantly being asked to learn a lot, and **it's a different kind of learning compared to what I'm used to in school...it has been a lot of learning on the job**, just because this is something that's, like, not really offered in any classes I could take."

— **Rikki**, a First-time intern developing infrastructure for the company's flagship product at an Applied Agentic AI Startup

"Now I'm learning, like, what C++ in the industry looks like, because it's different from school. Industry uses, like, the fancy pointers. **In school, we use raw pointers, so I finally learned the difference... a lot of it was getting used to industry-level coding standards.**"

— **Faye**, an Experienced intern developing a microservice at a Conglomerate

"I think **in school my code is a lot messier because...it's not, like, being thoroughly reviewed by anyone**. I just [need to pass] whatever automated tests they have, so I'm kind of just writing to make it work, whereas [in my internship] I'm kind of more writing to make it as polished as possible."

— **Ashley**, an Experienced intern developing a testing framework for AI agents at a Big Tech company

"Every internship I had, they might not be the best, but I feel like **I learned something I wouldn't learn in academia... I did not learn any, like, technical skills, but in terms of, like, life skills**, I learned a little bit about dealing with micromanagement, because maybe it will happen sometimes during my future career."

— **Koa**, an Experienced intern doing project management at a Financial Institution

"**I don't know if the school can really adequately prepare you...** I think it's a good thing that they encourage you to get internships in college. Because, yeah, that's just the only way you could learn how things go in the real world."

— **Cleo**, a First-time intern testing and automating an embedded software installer at a Government Contractor

"Realistically, **I don't think classes can be structured that ambiguously**. It would be hard for students to do well...there is that ambiguous structure to some of my classes. But in terms of, you know, meeting deadlines and expectations, that ambiguous structure isn't as clearly defined [in my internship]."

— **Lewis**, an Experienced intern automating AI modeling software at a Computer Chip Company

TAKEAWAY

Internships provided participants with an authentic CS learning experience that complemented their school-based learning.

DISCUSSION

We invite CS departments to explore ways to support students’ access to and success in internships through things like local business partnerships or credit-bearing companion courses.

ACKNOWLEDGMENTS

This material is based upon work supported by the National Science Foundation (NSF) CISE Graduate Fellowships under Grant No. 2313998. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the NSF. We also thank the Delta Lab for their valuable feedback.

[1] Caryn Thiên Ngán Tran, Max Kanwal, Kristin Fasiang, and Eleanor O'Rourke. 2025. Exploring Student-Perceived Dimensions of Authenticity in High School Computer Science. In *Proceedings of the 2025 ACM Conference on International Computing Education Research V.1*. ACM, Charlottesville USA, 93–108. doi:10.1145/3702652.3744229